

Orthogonal frequency division multiplexing phenomenology: radar technique combining genetic algorithm-based pulse design and energy detector for target recognition

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Abstract: This study investigates the benefit offered by the orthogonal frequency division multiplexing (OFDM) structure to enhance the recognition functionality in OFDM radar. The authors propose a novel phenomenology-based framework whereby the pulse is composed of several symbols each matched to a particular target and the recognition function in the receiver reduces to an energy detector. In this preliminary study, they make use of simple target models for which they have exact analytical expressions and apply genetic algorithm-based methods to design the symbols. They emphasise on the need to work within a less undeterministic environment and demonstrate the merit of the authors' technique with simulated results. Finally, they show how the integration of multiple echoes improves significantly the overall probability of correct classification despite additive white Gaussian noise.

1 Introduction

It is well-known that when illuminated by an electromagnetic wave the targets' radar cross-section (RCS) fluctuates with frequency and aspect angle [1]. This property and its consequences on the performance of the subsequent signal processing have been analysed by Swerling. He summarised his findings in the form of five Swerling models [2] where each model forms a pair composed of a distribution function of the RCS fluctuation and information about the rapidity of this fluctuation. The merit of these models is to provide analytical expressions for the probability of detection as a function of the average signal-to-noise ratio (SNR), which can be used by engineers to illustrate the merits of a radar design to potential clients.

Although RCS fluctuations with respect to frequency make the performance analysis more complex, we propose a radar system where we use the frequency signatures of targets to enhance the classification function directly in the measurement process. We support our approach with the following comments:

- The scattering mechanism can be modelled in the Fourier domain as a multiplication of the transmitted pulse spectrum with the target reflectivity spectrum [3].
- Parseval's theorem states that the total energy contained in the time domain signal summed across all of time t is equal to the total energy of its transform in the Fourier domain summed across all of its frequency components.
- Orthogonal frequency division multiplexing (OFDM) signals are generated in the frequency domain and their spectra can be tailored according to the needs within the bandwidth and energy constraints [4].
- The frequency signatures of targets are invariant over small aspect angle windows whose size is a measure of the decorrelation [1].

Essentially, this means that the transmitted pulse spectrum can be tailored to produce a received echo of high energy content provided that the target reflectivity spectrum is known over the frequencies of interest. It is this idea that we develop further to address a classification problem between several known targets. The procedure requires a controlled environment with a priori information on the targets at stake and the target aspect angle. In

particular, it is important that the target does not decorrelate within the bandwidth of interest during the observation time, as we shall see. Hence, our emphasis here is on civilian applications (e.g. security) for which such a controlled environment is possible, as opposed to military applications where the targets are not known.

Because the primary focus is on the scene, we suggest that this study falls within the theme of OFDM phenomenology [5]. The remaining of the paper is organised as follows. In Section 2, we review the prior work in relation to matched illumination methods for improved detection and recognition characteristics. Then, in Section 3, we introduce our technique. In particular, we elaborate on our contribution and present our simulation setup that includes a target modelling phase and a pulse design phase. In Section 4, we report our simulated results in an incremental fashion. We first assess the case when a single pulse and a deterministic target are used. Then we employ a stochastic target model to provide a statistical framework to the previous case. Next, we inspect the criterion of optimal frequency band for optimal recognition. Lastly, we investigate how the integration of multiple echoes can improve the overall recognition capability in noiseless conditions as well as when additive white Gaussian noise (AWGN) is added on the radar echoes. Section 5 concludes the paper.

2 Background

The field of cognitive radar (CR) has contributed significantly to the increased scope of waveform diversity [6–8]. The goal of CR is to provide the sensing system with the information about the propagation environment, input from external knowledge sources, system objectives and priorities so that the radar can be self-evolving. Haykin [9] introduced this close-loop process with a feedback running from the receiver to the transmitter. Using this method, the system is able to find the best waveform for the next illumination [9].

In this regard, several authors have considered the problem of finding the signal that optimises a particular task. Bell [10] developed a theoretical framework to find a waveform-receiver filter pair for the optimal detection of extended targets in additive noise. He introduced two different metrics (SNR or mutual information, MI), which are used depending if a deterministic

impulse response or a random target impulse response are considered. Pillai *et al.* [11] expanded the first part of this work (SNR metric) for the case where the target competes with signal-dependent interference. The authors developed an iterative algorithm to search for the transmit signal–receiver impulse response pair that converges satisfactorily for most values of input signal-to-interference-plus-noise ratio (SINR). They stressed that chirp-like pulses are most probably unacceptable solutions. Garren *et al.* [12] later adapted this algorithm to finite-dimensional vector spaces in order to facilitate a discrete-time implementation. Kay [13] proposed another extension of [10] to include the case of a Gaussian point target in Gaussian clutter. The merit of his approach is a simple analytical result which is guaranteed to be optimal. Lately, Romero *et al.* [14] expanded Bell's initial work to include the case where the target competes with signal-dependant interference with both metrics. Lastly, in [15], the authors proposed to design the transmit waveform as a weighted sum of basis sequences for the case of Gaussian point target in signal-dependent clutter and noise. Their idea is to maximise the detection performance with respect to the corresponding weighting factors so that the desired waveform is obtained immediately, while the aforementioned works derive the energy spectral density, which in turn infers a degradation loss as the time series are synthesised.

Besides the use of these so-called 'matched illumination-reception' techniques for optimising the target detection, various authors searched to apply the same concept to the problem of target classification. Garren *et al.* [12] built upon Guerci's work [16] to develop a theoretical framework that finds the optimal waveform for target identification discrimination when the scenario includes coloured noise and non-zero coloured clutter. Extensions of the identification problem are developed via a maximisation of the Mahalanobis distance between the echoes of the two target classes. As opposed to the first problem (target detection optimisation) there is no receiver filter involved. Another work [17] investigated further the problem while including aspect angle uncertainties for both cases of SINR maximisation and identification. In [6] both SNR and MI were used to adaptively modify the transmit waveforms for recognition of known targets when the additive noise is AWGN. It was shown that the integration of these waveform design techniques with sequential hypothesis testing could reduce the energy required for identification. In [18] the authors analysed further the recognition problem in the presence of clutter. They also demonstrated how their closed-loop framework and waveform design techniques could be applied to a multi-band transmission system where the system must adaptively select which band to use for any single transmission. More recently, Wei *et al.* investigated in [19] the design of phase-coded waveform for maximum recognition performance under detection constraints.

Lastly, Paichard *et al.* investigated in [20] the benefit to transmit OFDM pulses to acquire the reflectivity spectra of time-varying targets. The relevance of this study shall become clear as we discuss target modelling in Section 3.3. In particular, the use of similar systems like their proposed HYperfrequency CAMera (HYCAM) may permit to acquire reflectivity spectra from target of interest on the fly.

3 Technique

Our technique seeks to propose a simple yet reliable framework to support the recognition task when the radar has to choose among several targets characterised by their deterministic impulse responses, or in the Fourier domain their reflectivity spectra. We follow the phenomenological approach first introduced by Huynen [21] whereby most of the focus shall be on the phenomenon under interest. Although it is not our goal to develop a bio-inspired radar mechanism, our approach can be used in that scope [22].

3.1 Contribution

As opposed to any of the aforementioned works, our technique relies on the OFDM structure for the transmitted pulse. The use of this

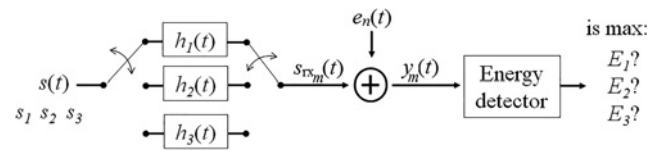


Fig. 1 Simulation setup of the matched-filterless recognition concept

technology is motivated by the nature of the OFDM and in particular the fact that the time domain pulse is efficiently synthesised from the spectrum. To our best knowledge, the concept whereby the recognition task is solely based on the use of a single pulse (possibly transmitted multiple times) coupled with an energy detector is novel. Furthermore, we are not trying to find an analytical expression for the optimal radar signal used for recognition, as we make use of an iterative search method based on genetic algorithms (GA). This procedure is fast, allows a large number of signals to be generated and can easily include additional constraints [4].

3.2 Simulation setup

Our concept is better described by the block diagram given in Fig. 1. In this preliminary study, we look at the case of three targets. Hence, our transmitted signal $s(t)$ is composed of three sub-pulses s_1 , s_2 and s_3 which we refer to as OFDM symbols 1, 2 and 3. The pulse interacts with one of the three targets in the scene, characterised by its impulse response h_m , where $m=1, 2$ or 3 . In the time domain, this process is modelled by a convolution between the transmitted signal and the deterministic target impulse response. AWGN adds onto the echo and the energy detector then simply evaluates the amount of energy on each of the sub-echoes. Classification of the target is done upon that information. The full process employed to generate the OFDM symbols is summarised as follows:

- (i) Characterise N_T target models
- (ii) Calculate N_T reflectivity spectra
- (iii) Generate N_T masks
- (iv) Design N_T symbols
- (v) Concatenate all N_T symbols to compose the pulse

The details of these steps are illustrated in the following sections.

3.3 Target modelling

In our experiment, we use target models constructed from several perfectly electrically conducting (PEC) spheres such that we can apply the Mie theory. In fact, spheres are the only objects for which there exists a closed-form expression for the backscattered electrical field. Together with the superposition principle, we obtain analytical expressions for the RCS as a function of frequency. A similar experimental setup was proposed in [23], where several automatic target recognition (ATR) algorithms are tested on synthetic radar data for recognition purposes. In this work, the emphasis was on the processing side.

(i) *Mie scattering*: The Mie scattering is in fact the Mie solution to the Maxwell's equations named after Gustav Mie that describes the scattering of electromagnetic radiation by a sphere. The solution takes the form of an infinite series [24]. In our experiment, we consider spheres with a size comparable to the wavelength for the range of frequencies of interest so that the RCS varies in the resonance or the Mie region. It is this fluctuating behaviour that we intend to use to distinguish between the multiple targets.

(ii) *Models*: The simulation environment assumes that the radar is located 10 km away from the target, hence its coordinates are $(-10 \text{ km}, 0)$ and the centre of the reference system lies in the vicinity of the target. Each sphere is characterised by the coordinates of its centre (x_s, y_s) as well as its radius a_s . The target

models used in our experiment are presented in Figs. 2a–c. To simulate different targets, we vary the number of spheres, their sizes and positions. Different setups may consider other models. Our intention here is to generate a preliminary set of synthetic targets upon which we can test our technique.

Arbitrarily, we consider a bandwidth of 30 MHz between 1 and 1.03 GHz. The objective is to have a bandwidth large enough so that the RCS of all three targets behave differently over the band. Figs. 2d–f show the targets' RCSs over an extended band between 0.9 and 1.1 GHz. The part within the red rectangle corresponds to the 30 MHz bandwidth.

(iii) *RCS normalisation*: Before manipulating further the RCS profiles, we must normalise them so that they become comparable.

The normalisation is performed on the reflectivity spectra following the method proposed in [7, chap. 14], viz., in the Fourier domain, the reflected signal resulting from a flat spectrum OFDM pulse of unit energy $\mathbf{s}$ shall have unit average power. The normalised RCS profiles are finally given in Fig. 3a, while the phase of the reflectivity spectra is given in Fig. 3b. Importantly, we observe that all three targets' RCS exhibit peaks at different frequencies.

(iv) *Target impulse response*: For each target model, the impulse response in vector format, $\mathbf{h}_t$ is obtained by taking the inverse discrete Fourier transform (IDFT) of the target reflectivity spectrum, hence $\mathbf{h}_t = \text{IDFT}(\mathbf{h}_f)$. To maintain reasonable values in our simulations, we normalise each vector $\mathbf{h}_t$ so that they have unit energy. We then use these normalised vectors to generate the received echoes. The normalised impulse response of Targets 1–3 is given in Figs. 2g–i. While the peak appears at 10 km for both Targets 2 and 3, its position is shifted by one sample, hence at

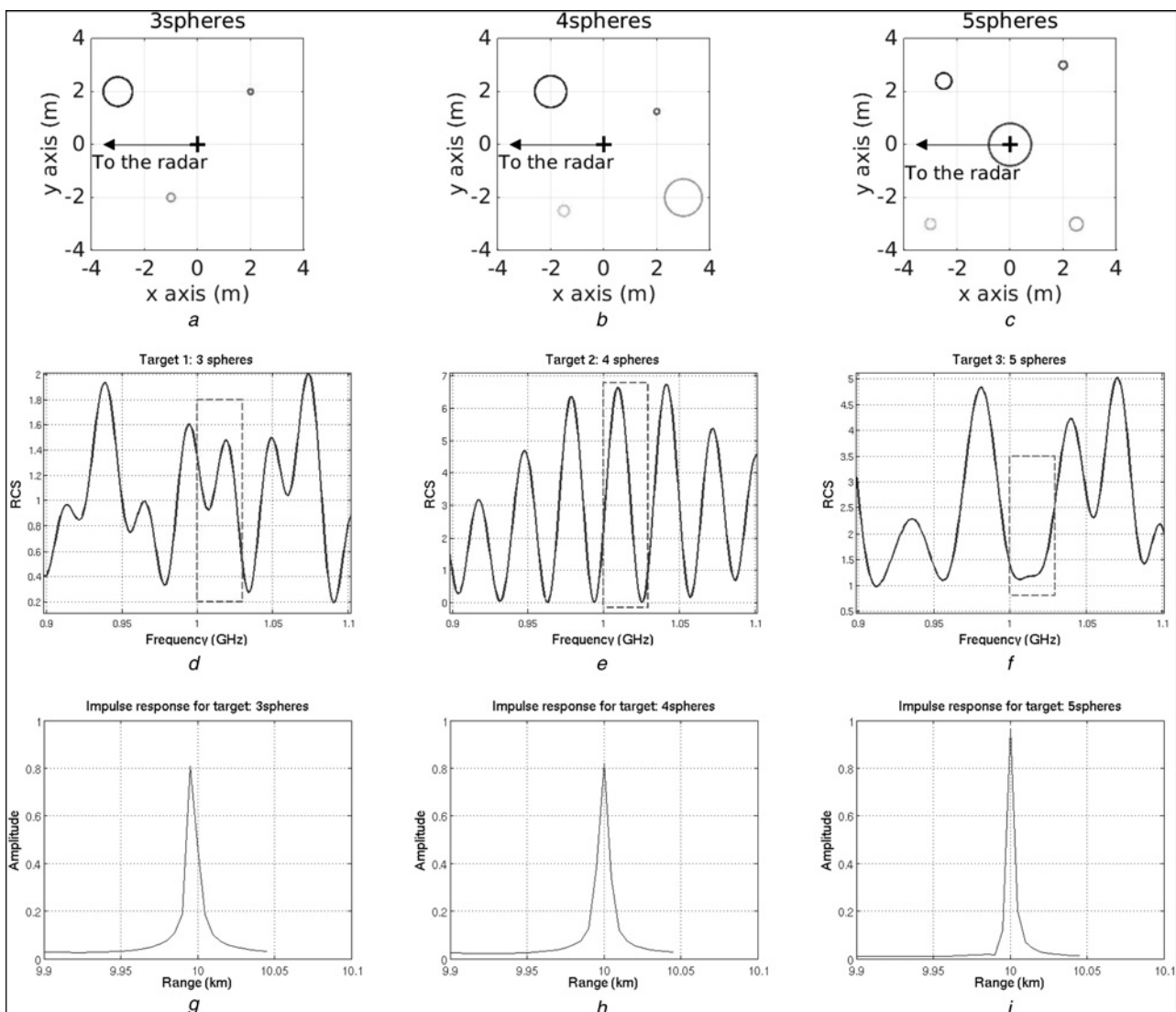


Fig. 2 Geometrical model for the targets considered in the experiment Coordinates of the centres and the radii a_s are as follows: (a) $(-3, 2)$, $(-1, -2)$, $(2, 2)$ and $a_s = 0.55$ m, $a_s = 0.15$ m, $a_s = 0.1$ m, respectively, (b) $(-2, 2)$, $(-1.5, -2.5)$, $(2, 1.25)$, $(3, -2)$ and $a_s = 0.6$ m, $a_s = 0.2$ m, $a_s = 0.1$ m, $a_s = 0.7$ m, respectively and (c) $(-2.5, 2.4)$, $(-3, -3)$, $(2, 3)$, $(2.5, -3)$, $(0, 0)$ and $a_s = 0.3$ m, $a_s = 0.2$ m, $a_s = 0.15$ m, $a_s = 0.25$ m, $a_s = 0.8$ m, respectively. (d)–(f) RCS of the three targets considered in the experiment over a frequency band comprised between 0.9 and 1.1 GHz. (g)–(i) Impulse responses for all three targets

- a Target 1
- b Target 2
- c Target 3
- d RCS $|h_1(f)|^2$
- e RCS $|h_2(f)|^2$
- f RCS $|h_3(f)|^2$
- g Impulse response for Target 1
- h Impulse response for Target 2
- i Impulse response for Target 3

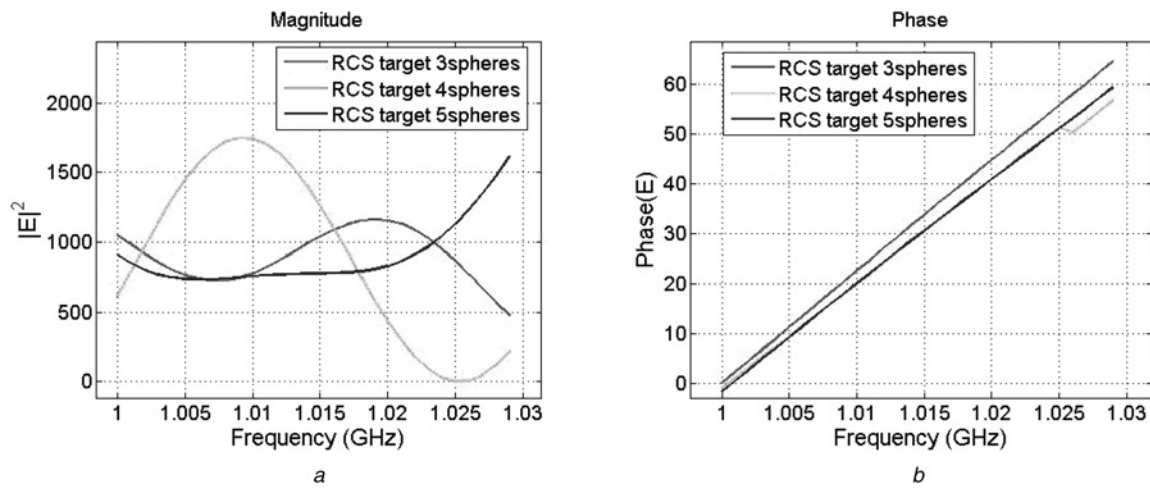


Fig. 3 Normalised RCS and phase of the reflectivity spectra of all three targets within the 30 MHz bandwidth of interest

a Normalised RCS
b Phase of the reflectivity spectra

9995 m in the case of Target 1. This is in agreement with the geometrical model given in Fig. 2a.

3.4 Randomisation

To establish a proof of concept, we suggest to use a Monte Carlo approach on a large data set. Essentially, we generate M reflectivity spectra from M realisations of the target when a slight change on the spheres' coordinates and radii is applied. For each realisation of the target, any of these three characteristics x_{sm} , y_{sm} and r_{sm} are generated from normal distributions with means x_s , y_s and r_s , respectively, and standard deviation $\sigma_s = 0.01$ m. We

illustrate this approach with Figs. 4a–c for Target 1. We used $M = 5000$. With $\sigma = 0.01$ m, 99.8% of the points are comprised within $6\sigma \simeq 1\% \cdot \delta R$, where $\delta R = c/2B = 5$ m is the range resolution. These stochastic models are used to reflect the uncertainties over the actual target. Importantly, in real applications, the controlled environment is required to guarantee that the target in the scene matches the model considered. Indeed, we seek to reduce the risk of decorrelation of the actual reflectivity spectrum as compared to the simulated reflectivity spectrum.

The result of the stochastic models is a set of reflectivity spectra, which we can use to build a set of histograms for the frequencies of interest. Each histogram is a measure of the reflectivity spectrum for a particular frequency. When normalised, we can interchangeably

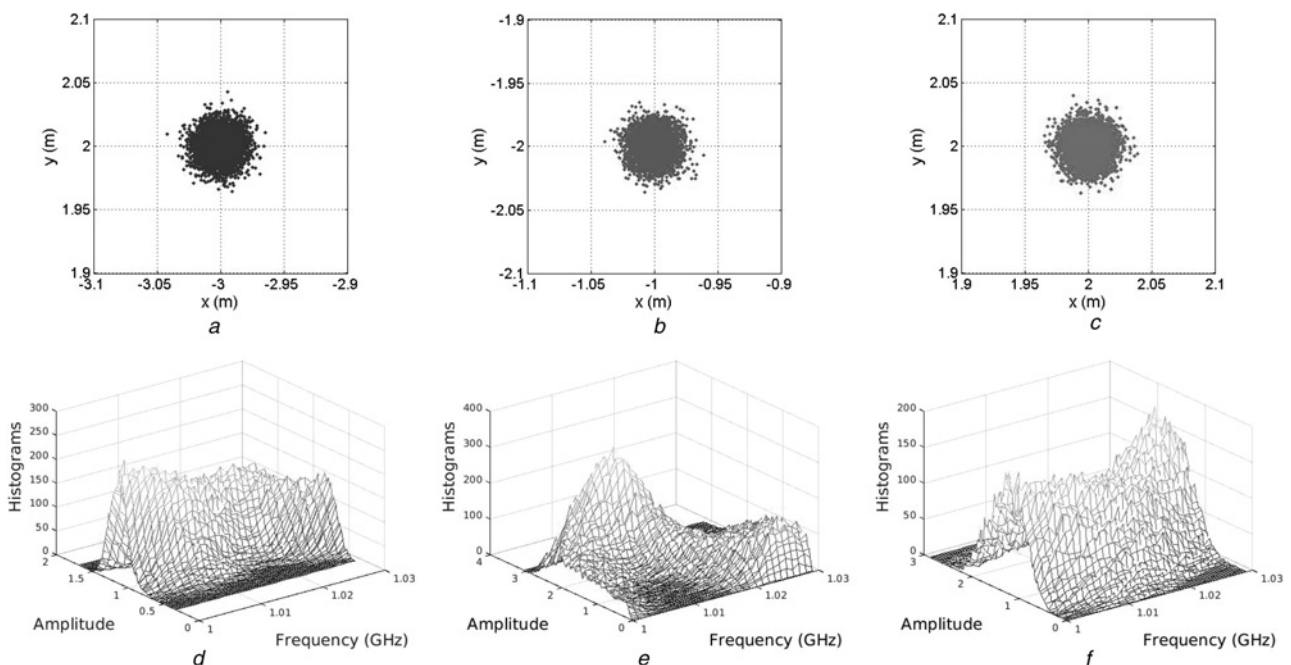


Fig. 4 (a)–(c) Effect of the randomisation on the position of the centre of the three spheres for Target 1. We use a normal distribution centred on the original values with a standard deviation of 0.01 m. (d)–(f) Set of histograms of the reflectivity spectrum for the three target models as a result of the randomisation

a Centres for sphere 1
b Centres for sphere 2
c Centres for sphere 3
d Set of histograms for Target 1
e Set of histograms for Target 2
f Set of histograms for Target 3

refer to these histograms as probability density functions (PDFs). The sets of histograms for the three target models are shown in Figs. 4d–f. For each target, we are able to retrieve the trend of the RCS plot given in Fig. 3a. These plots are important for they indicate the degree of correlation or decorrelation present in the set of target models. In fact, the sharper these histograms, the stronger the correlation. Hence, the variations seen in the histograms infer that the correlation level is not equal along the frequency axis. For example, in Fig. 4e, the level of correlation is strong in the first 10 MHz from 1 to 1.01 GHz and it reduces in the next 20 MHz of the bandwidth. In contrast, the level of correlation for Target 3 remains high throughout the band as shown in Fig. 4f.

3.5 Pulse design

As evoked in Section 3.1, our concept uses the OFDM technology. The incentive to use OFDM is twofold: first, the signal is fully defined by its discrete spectrum vector. Second, the discrete time domain signal is easily synthesised when applying an inverse fast Fourier transform (IFFT) algorithm on the discrete spectrum vector. Hence, provided that we find optimal discrete spectra, the time domain signals are generated at no loss. In this section, we first discuss the generation of our so-called masks which we use to search for optimal discrete time domain signals. We remark that the peak-to-mean-envelope power ratio (PMEPR) constraint, particular to OFDM, can be further optimised without impinging the main objective of classification [4].

3.5.1 Generation of the masks: The generation of the masks is an important step in the design of the pulse as shown earlier. Intuitively, the mask used for a given target would comprise of the parts of its reflectivity spectrum that exceed the reflectivity spectra of any other targets, in a normalised sense. If we were to design the OFDM symbol to match the mask, we would excite the target of interest while refraining from exciting the other. Repeating the same procedure for each of the N_T targets under consideration would result in N_T OFDM symbols, each one being matched to only one target. Concatenating those N_T symbols to form the radar pulse, the signal received after the pulse bounces off any of those N_T targets shall present the highest energy level within the corresponding sub-echo. It is this principle introduced in Fig. 1 that we develop in greater detail now. The mechanism employed to generate the masks is presented in the following algorithm in Fig. 5.

Figs. 6a–f illustrate the procedure for the case where the three targets from Fig. 2 are considered. For each target, the difference in reflectivity spectrum with respect to the other two targets and

the resulting mean are given. The masks that result from our algorithm in Fig. 5 are also plotted. As expected, the non-constant parts coincide with the frequency segments when the mean profile is higher than the threshold α_{th} . In our procedure, we chose $\alpha_{th} = 5$ to guarantee that the mask reflects the peaks of this mean profile and the default value is chosen as $\varepsilon = 1$. This last value is arbitrary. We reckon that it shall be small enough in comparison to α_{th} to enhance the peak. We observe in Fig. 6b that the differences of the reflectivity spectra between Targets 2 and 1 and Targets 2 and 3 are strong over the same portion of the spectrum. This is the ideal case as shown in Section 4. When more than three targets are considered, the bandwidth shall be chosen to ensure high differences in RCS for all targets. In this case, we believe that the bandwidth needs to be wide enough and reckon that the use of a restricted bandwidth may be harmful.

3.5.2 Sub-pulses optimisation: As indicated in Section 3.2, the next step is to design the pulse. More precisely, our task is to search for the N_T optimal OFDM symbols (also called sub-pulses). We define our optimality criterion as follows: the optimal OFDM symbol shall, after it has been normalised, produce an echo of maximum energy when bouncing off a hypothetical target itself characterised by the normalised mask generated earlier. In mathematical terms, our optimisation problem reduces in finding the weights w_n that satisfy the following equation

$$\arg \max_{w_n} |\mathbf{z}_n^T \mathbf{m}_{n \text{ norm}}|^2, \quad \text{s.t.} \quad \begin{cases} w_n \geq v_l \\ w_n \leq v_u \end{cases} \quad (1)$$

where $\mathbf{m}_{n \text{ norm}}$ is the mask vector after normalisation and $\mathbf{z}_n$ is the normalised complex spectrum of the OFDM symbol resulting from (2)

$$\mathbf{z}_n = 1/\sqrt{N_s} \sum w_n^2 \mathbf{z}_n, \quad (2)$$

where $\mathbf{z}_n = [z_1 \dots z_{N_s}]^T$ and $z_n = w_n a_n$ are complex numbers such that $|z_n| = w_n$, $\arg(z_n) = \arg(a_n)$. T denotes the matrix transposition. The length of the OFDM spectrum vector is N_s and corresponds to the number of subcarriers used. Since the energy is considered in this optimisation procedure, only the weights are relevant. In fact, the arguments of z_n can be used to solve the other problem of the peak-to-mean-envelope-power ratio in the same way it is done in [4]. Note that our constraints on w_n in (1), namely v_l and v_u are introduced to limit the dynamic range of the weights. In the case, we would impose a wideband transmission the dynamic range shall be small. Otherwise, it can be high. In the following,

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1: for target model  $n=1$  to  $N_T$  do
2:   Calculate the mean magnitude and phase of the reflectivity spectrum from target  $n$ ,  $\bar{\mathbf{h}}_n$ .
3:   Normalise the reflectivity spectrum  $\mathbf{h}_{n \text{ norm}}$ .
4:   for all other  $N_T-1$  target models  $nn$  do
5:     Calculate the mean magnitude and phase of the reflectivity spectrum from target  $nn$ ,  $\bar{\mathbf{h}}_{nn}$ .
6:     Normalise the reflectivity spectrum  $\mathbf{h}_{nn \text{ norm}}$ .
7:     Generate the difference of the reflectivity spectra  $|\Delta \mathbf{h}_{n,nn}| = |\mathbf{h}_{n \text{ norm}}| - |\mathbf{h}_{nn \text{ norm}}|$  and,
        $\arg(\Delta \mathbf{h}_{n,nn}) = \arg(\mathbf{h}_{n \text{ norm}}) - \arg(\mathbf{h}_{nn \text{ norm}})$ .
8:   end for
9:   Calculate the mean magnitude, viz.,
        $|\bar{\Delta \mathbf{h}}_n| = 1/(N_T-1) \sum_{nn=1}^{N_T-1} |\Delta \mathbf{h}_{n,nn}|$ .
10:  Calculate the mean phase, viz.,
        $\arg(\bar{\Delta \mathbf{h}}_n) = 1/(N_T-1) \sum_{nn=1}^{N_T-1} \arg(\Delta \mathbf{h}_{n,nn})$ .
11:  Fix a threshold  $\alpha_{th}$ 
12:  Find the frequency indices  $k_i$  where  $|\bar{\Delta \mathbf{h}}_n| > \alpha_{th}$ .
13:  Set the mask corresponding to target model  $n$  equal to a default value  $\varepsilon$ :  $\mathbf{m}_n = \varepsilon[1 \dots 1]$ .
14:  For the indices  $k_i$  set the magnitude of the mask equal to the mean magnitude of the difference reflectivity
       spectra:  $|\mathbf{m}_n[k_i]| = |\bar{\Delta \mathbf{h}}_n[k_i]|$ .
15:  For the indices  $k_i$  set the phase of the mask equal to the mean phase of the difference reflectivity
       spectra:  $\arg(\mathbf{m}_n[k_i]) = \arg(\bar{\Delta \mathbf{h}}_n[k_i])$ .
16:  Normalise the mask  $\mathbf{m}_{n \text{ norm}}$ .
17: end for

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Fig. 5 Generation of the masks

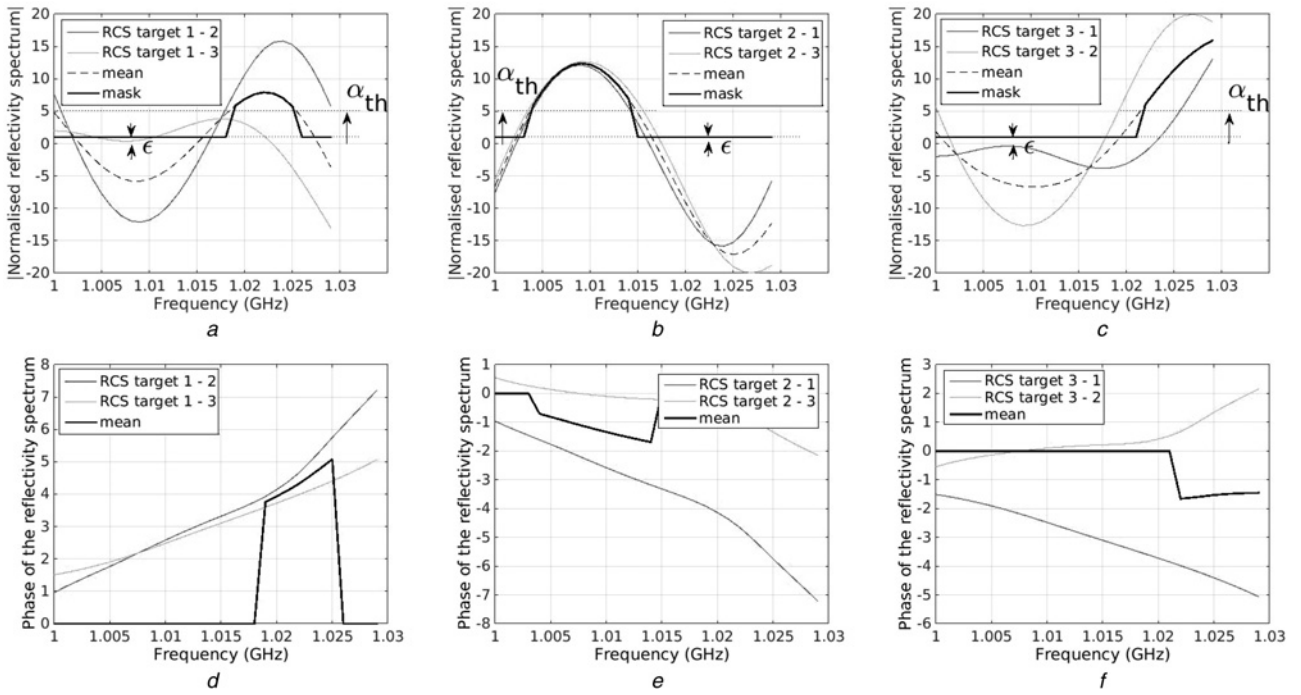


Fig. 6 Generation of the masks for the three targets case. Magnitudes (a)–(c) and phases (d)–(f)

a $|m_1|$
b $|m_2|$
c $|m_3|$
d $\arg(m_1)$
e $\arg(m_2)$
f $\arg(m_3)$

we present simulation results obtained for different values of v_l when v_u is fixed. The orthogonality principle [25] ruling the OFDM structure infers that the subcarrier spacing Δf and the symbol (or bit) duration t_b are linked according to $\Delta f = 1/t_b$. Hence fixing the symbol duration fixes the subcarrier spacing and then fixing the bandwidth B fixes the number of subcarriers N_s . In the present case, we use $B = 30$ MHz and $t_b = 1 \mu s$ so that $\Delta f = 1$ MHz and $N_s = 30$ subcarriers.

This optimisation problem suggests a heuristic search, hence we propose to apply a genetic algorithm based method which has proven to be well suited to similar problems [4, 26]. In fact, we propose to apply the continuous genetic algorithm since the variables are not naturally quantised. The vast amount of literature available on the topic [27, 28], therefore we restrict our discussion to the specificities of our application.

• **Cost function:** The cost function which we use in our GA is the same scalar product as in (1) to the difference that we introduce a minus sign in order to work out a minimisation rather than a maximisation

$$\text{cost} = f(\text{chromosome}) = -|\mathbf{z}_n^T \cdot \mathbf{m}_n|^2. \quad (3)$$

• **Variables:** As indicated in the cost function, the variables are the N_s pairs of weights and arguments $[w_n, \arg(a_n)]$ resulting in $2N_s$ variables. In fact, as we already explained only the weights w_n impact the cost function, thus it is possible to reduce the variables to be the N_s weight values. In our simulations, we consider the first case of $2N_s$ variables.

• **GA parameters:** The constraints defined in (1) give the possible values for the weights and the phase codes a_n can take any value around the unit complex circle. We consider running our simulations over 1000 iterations, we use a population of 20 ‘chromosomes’, take a mutation rate of 0.2 and a selection rate of 0.5. Of course, different sets of parameters are possible, but it is not our intention to address the problem of optimising our algorithm. Our aim is to show that this simple GA based technique can produce suitable results.

• **Initial population:** The initial population is generated at random. In other words, 20 vectors of length $2N_s$ each are generated. The weights follow a uniform distribution within $[v_l, v_u]$. Likewise, the phases of the phase codes follow a uniform distribution within $[0, 2\pi]$.

• **Convergence check:** The convergence check is based on the stopping criteria defined earlier. Hence, we stop once the number of iterations has reached the end.

• **Final population:** The final population has the same size as the initial population. We select the element with the best cost and use it to generate our optimal OFDM symbol.

In practice, we include in the initial population a chromosome whose weight and phase vectors are identical to the mask, provided some scaling factor for the weight vector so that its values remain within the bounds $[v_l, v_u]$. Intuitively, we expect from vectors whose behaviour is similar to the mask to have favourite characteristics as far as the energy of the received echo is concerned. In Figs. 7a–c, we tracked the cost function of the best chromosome in the population versus the generation index. The decreasing trend of the curve attests the gain produced by our GA. We considered three cases, for three different values of v_l , while $v_u = 10$. Namely, we used $v_l = 0.01, 1$ and 5 . Some optimal solutions after 1000 iterations for the first target mask are presented in Figs. 7d–f. When the dynamic range reduces as in Fig. 7f the optimal weight vector spreads its energy over the entire peak.

Lastly, the time domain sub-pulses s_n are formed after applying an IDFT on the complex vectors $\mathbf{z}_n$ and the pulse is composed by concatenating the three optimal sub-pulses.

4 Statistical analysis

In this section, we evaluate our concept presented in Fig. 1. We first report on our preliminary results where each target is modelled by a single deterministic impulse response. We then introduce the results

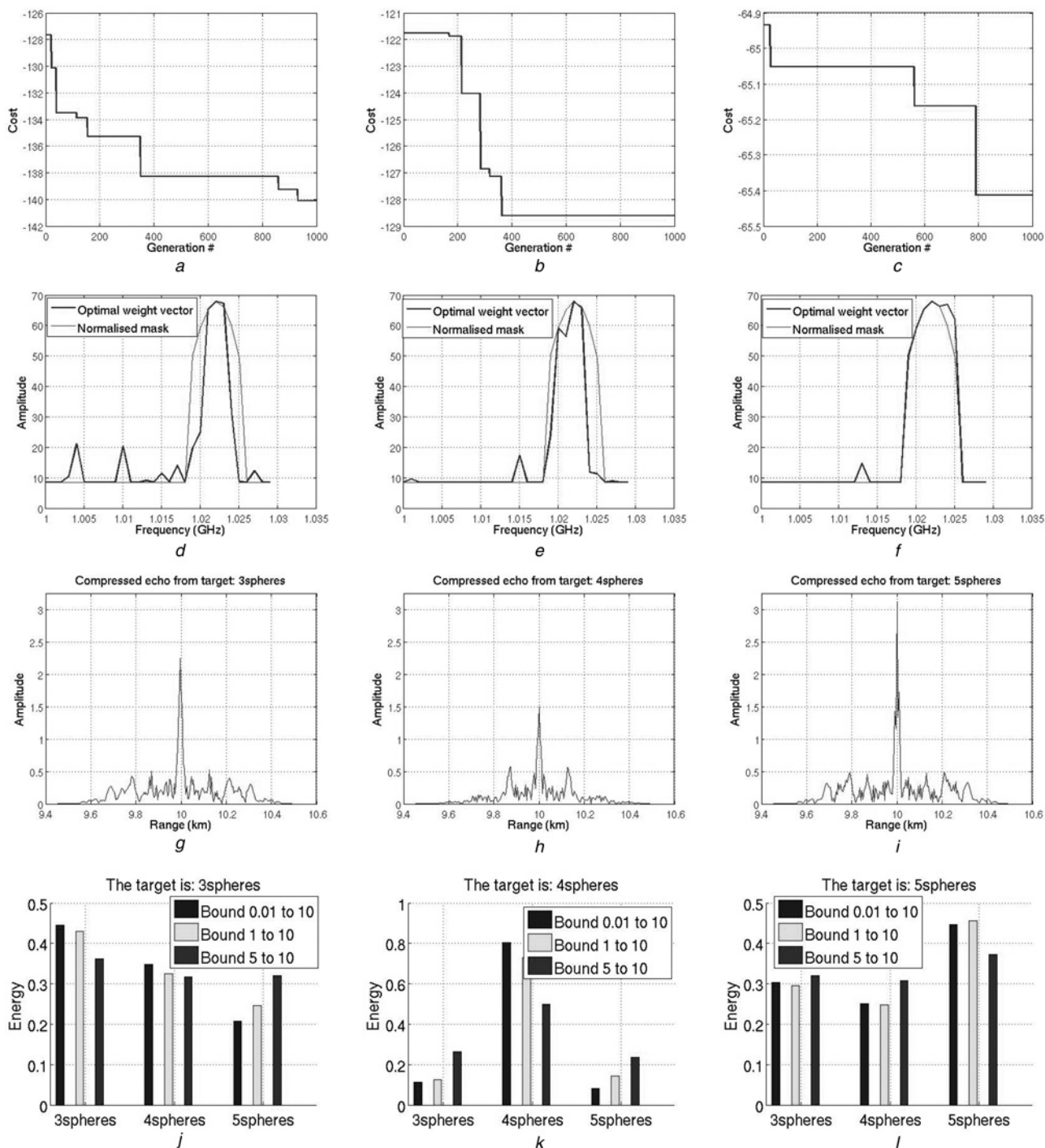


Fig. 7 (a)–(c) Cost value as a function of the generation index for Target 1 and different values of v_l . (d)–(f) Mask and corresponding weight vector for different values of v_l . Note that we used a scaled and shifted version of the weight vector to plot both quantities on the same graph. Our interest lies in the overall allure of the weight vector in comparison to the mask. (g)–(i) Compressed echoes for either Targets 1, 2 or 3. Asymmetric allure of the curve results from the extended target behaviour, and as such is different from the simple autocorrelation function of the transmitted pulse. (j)–(l) Energy content in the received sub-echoes for different values of v_l

a Cost for bounds 0.01–10
 b Cost for bounds 1–10
 c Cost for bounds 5–10
 d Lower bound: 0.01
 e Lower bound: 1
 f Lower bound: 5
 g Compressed echo from Target 1
 h Compressed echo from Target 2
 i Compressed echo from Target 3
 j When Target 1 is present
 k When Target 2 is present
 l When Target 3 is present

obtained from Monte Carlo simulations. For each realisation, the target has a slightly different geometrical model, which infers a slightly different reflectivity spectrum as explained in Fig. 4. This approach enables us to generate statistical results for the recognition problem. We then propose to introduce a search technique to find the optimal band in the spectrum as far as recognition is concerned. We compare the results obtained for two cases and discuss the tradeoffs. Finally, we inspect the benefit of integrating several pulses before assessing the energy content of the various sub-echoes. We show that this technique brings tremendous improvements even when low SNR conditions are considered.

4.1 Preliminary results

4.1.1 Unitary test: In fact, prior to the energy detector presented in Fig. 1, one needs to localise the beginning of the pulse. To that end, we suggest making use of a matched-filter. Hence, we filter the received echo with a replica of the transmitted pulse and the peak in the output is used as an indicator of the first pulse sample. Figs. 7g–i illustrate the compressed echoes for all three targets. The difference in the peak amplitude results from the different values of the three target RCSs despite the normalisation method presented in Section 3.3(iii). In particular, we observe that the peak is positioned at 10 km for both Targets 2 and 3, while it appears on the previous sample, 9.995 km for Target 1. As pointed out before, this is in line with the geometrical target models for Target 1 presents its main contributor at a range of 9.997 km from the radar, which is closer from the 9.995 km range bin than the 10 km one.

Following our prior preparation, we repeat the experiment for each of the three pulses that we generated with different values of v_l . The energy values E_1 , E_2 and E_3 are assessed and our results are given in Figs. 7j–l. For the sake of clarity, prior to summing the samples, we normalise the received echo so that its energy is 1. Hence, 100% of the energy is distributed over all three sub-echoes. We can clearly expect to get the highest energy content in the symbol corresponding to the target in the scene. This intuition is verified for all three pulses designed with $v_l = \{0.01, 1, 5\}$. However, we observe that the smaller v_l the better the discrimination. As a matter of fact, from a recognition point of view, it is optimal to transmit most of the energy on the few subcarriers where the magnitude of the mask is strong as it is the case in Fig. 7d.

The different behaviour of the histogram for Targets 1 and 3 as opposed to Target 2 is to be expected if the amplitude of the mask in Fig. 6b is compared to the three target RCSs shown in Fig. 3a. In the case of Target 2, it is clear that the mask puts most of the energy on those frequencies where the RCS from Target 2 exceeds significantly the RCS from both Targets 1 and 3. Hence, we can anticipate the result whereby the second symbol will excite Target 2 very effectively. Although symbols 1 and 3 are exciting Targets 1 and 3 sufficiently to give rise to higher values for E_1 and E_3 , respectively, the other values are not significantly lower. Obviously, the case from Target 2 reveals the ideal configuration as it infers an optimal recognition pattern. As a matter of fact, we discuss later the possibility to select an optimal bandwidth that could maximise the recognition pattern in all three cases, not only one as it is presently the case.

4.1.2 Monte Carlo simulation: To characterise statistically our concept, we run a Monte Carlo simulation where we make use of the

Table 1 Confusion matrix resulting from the stochastic target models with $\sigma = 0.01$ m

		Predicted target		
		Target 1, %	Target 2, %	Target 3, %
actual target	target 1	63.3	36.2	0.5
	target 2	5.3	94.2	0.5
	target 3	9.3	30.1	60.6

stochastic RCS models as generated in Section 3.4. For each realisation, the target is described by a slightly different RCS curve. In the end, we observe the distribution of the energy level in all three symbols, but more importantly we check whether the appropriate sub-echo is the most energetic, in which case we declare that the recognition has been successful. This approach enables us to build a confusion matrix which we present in Table 1.

The confusion matrix is one of the most adopted ways of representing the results from an ATR exercise. Details concerning the confusion matrix derivation can be found in a paper by Mishra and Mulgrew [29]. For the configuration under consideration (stochastic target models with standard deviation $\sigma = 0.01$ m for both the sphere positions and radii), Table 1 confirms the outstanding performance (94.2%) as far as the recognition of Target 2 when Target 2 is present in the scene is concerned. For the other two cases, our conclusion is mitigated since high probabilities of misclassifications P_{mc_i} are observed. The latter is calculated following the equations in [29]. Likewise, the probabilities of false alarm P_{fa_i} for a given target i are calculated as well as the overall probability of correct classification P_{cc} . This information, extracted from the confusion matrix, is summarised in Table 2.

In short, these preliminary results demonstrate the capacity of our technique to classify between several targets, when their RCSs are known although clearly, improvements to reduce the false alarm probabilities and increase the overall probability of correct classification are needed. These results also stress the role of the frequency window in the generation of optimal masks. In that respect, we suggest to consider an additional step where we first search for an optimal frequency window.

4.2 Optimal frequency band search

The idea is to see whether a more judicious choice of the frequency window could improve the recognition performance. Note that the bandwidth remains the same, 30 MHz. In the present case, we suggest to run our search arbitrarily within the interval [1–2 GHz]. From our aforementioned results, we understand that an optimal configuration would be achieved if the RCSs would follow the pattern given in Fig. 8a. Even though this ideal configuration may not happen within the band of interest, our intention is to converge towards the best solution. We summarised our search technique in the following algorithm in Fig. 9.

Figs. 8b and c used to illustrate the behaviour of the algorithm. In our configuration where the threshold is defined as the mean of all three RCSs, the first window is not selected by the algorithm because two peaks are present in the first interval. The vertical lines depict the frequency grid. Conversely, Fig. 8c shows the RCSs corresponding to one of the optimal solutions. We see that the peaks are well separated so that the maximum for one target RCS will certainly coincide with very low values for the other target RCSs. The peak values are not perfectly equivalent, however the spread remains acceptable based on the threshold that we selected. In the following section, we compare the results obtained as we work with the *default* [1–1.03 GHz] or this *optimal* frequency window [1.45–1.48 GHz]. In other configurations (different target models and different number of targets) this mechanism shall be adapted.

4.3 Results and comparisons with one pulse

To understand the ‘benefit’ of the optimal window we generate RCS data from the stochastic geometrical models as shown in Fig. 4 for

Table 2 Information extracted from the confusion matrix in Table 1

Target i	P_{mc_i} , %	P_{fa_i} , %	P_{cc}
1	36.7	7.3	72.7%
2	5.8	33.2	
3	39.4	0.5	

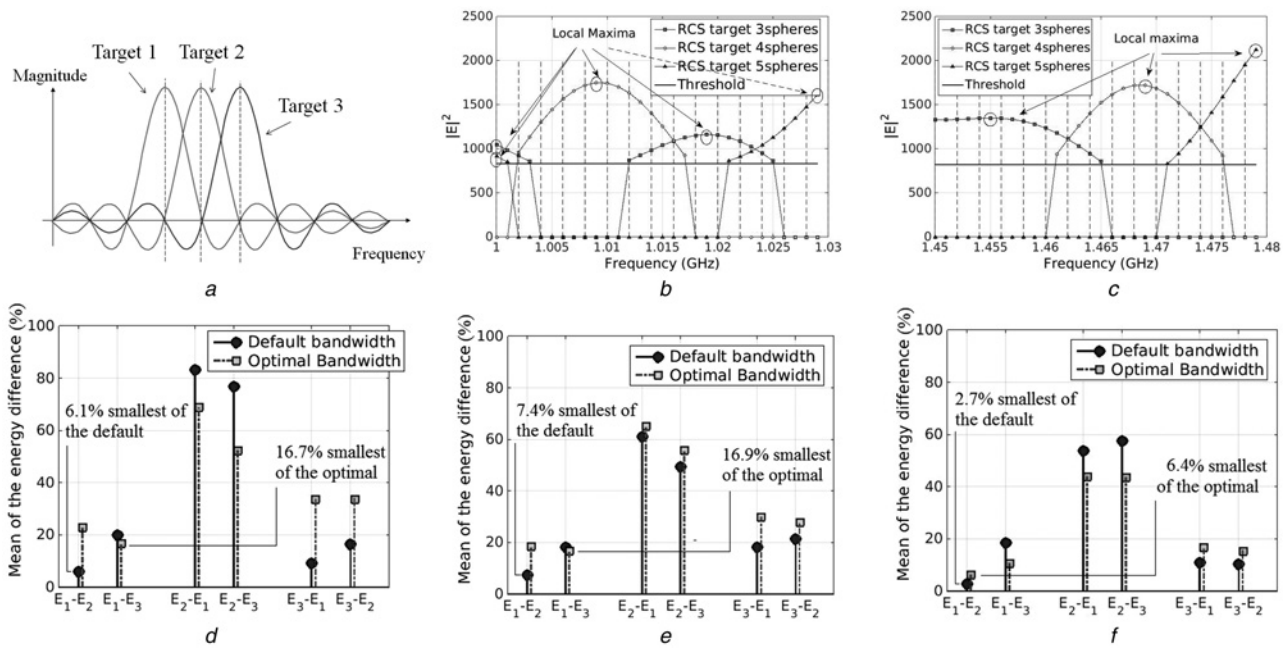


Fig. 8 (a) Ideal situation for the targets' RCSs to provide optimal recognition characteristics. Just like the OFDM structure, we want that the peak for one RCS coincides with low values for the other. (b) and (c) RCS curves illustrate the behaviour of this algorithm for two different frequency windows, (b) the first window starting at 1 GHz and (c) one of the optimal windows resulting from the search and starting at 1.45 GHz. (d)–(f) compare the energy differences for the two bandwidth cases

a Ideal RCSs
 b First window at 1 GHz
 c Optimal window at 1.45 GHz
 d $\sigma = 0.001$ m
 e $\sigma = 0.005$ m
 f $\sigma = 0.01$ m

different values of the standard deviation σ . We consider the next three values, 0.001, 0.005 and 0.01 m; essentially, the smaller σ , the less decorrelation throughout the 30 MHz band. In fact, as we generate three-dimensional (3D) plots for the optimal window case at 1.45 GHz as shown in Figs. 4d–f we observe a stronger decorrelation at $\sigma = 0.01$ m, which we explain by the higher frequencies covered. To present our results we follow two approaches.

In the first approach, for every realisation, we assess the difference in energy between the sub-echo corresponding to the target in the scene and the other two (e.g. $E_1 - E_2$ and $E_1 - E_3$, when Target 1 is in the scene). Then we average out to get the mean value which we plot in Figs. 8d–f. This approach gives us an idea of the margin for the recognition. Our aim with the *optimal* frequency window was to improve the recognition properties for the cases

where either Target 1 or 3 are in the scene. Figs. 8d–f confirm that those margins have been significantly improved. In particular, with $\sigma = 0.001$ m, the minimum margin increased from 6.1 ($E_1 - E_2$) to 16.7% ($E_1 - E_3$); with $\sigma = 0.005$ m, the minimum margin increased from 7.4 ($E_1 - E_2$) to 16.9% ($E_1 - E_3$); with $\sigma = 0.01$ m, the minimum margin increased from 2.7 ($E_1 - E_2$) to 6.4% ($E_1 - E_2$).

Our second approach consists of working out the confusion matrices. To reduce the computational load, we implemented Monte Carlo simulations based on 1000 iterations rather than 5000 as shown in Section 4.1.1. Comparing both cases of *default* and *optimal* windows we see that almost 100% of recognition rate is achieved. However, as σ increases, the results for the *optimal* case will degrade more strongly. For instance, with $\sigma = 0.01$ m, the probability of correct classification of Target 1 is 47.7% in the *optimal* case as opposed to 63.7% in the *default* case.

```

1: Initial frequency band = [1 GHz, 1.03 GHz]
2: for each frequency band within [1 GHz, 2 GHz] do
3:   Define a frequency grid within that band (e.g. 1 GHz, 1.002 GHz, 1.004 GHz etc.)
4:   for each target do
5:     Identify the portion(s) of RCS above a certain threshold (e.g. it can be the mean value of all RCSs)
6:     For every portion, identify the frequency corresponding to the peak
7:     Identify the interval of the frequency grid where the peak occurs
8:     Store the peak value (if several peaks, store the maximum value)
9:   end for
10:  if No more than one peak is observed in any interval of the frequency grid then
11:    if The separation between two consecutive peaks is large enough then
12:      if The maxima of the RCSs are roughly equivalent then
13:        Tick the current frequency band
14:      end if
15:    end if
16:  end if
17: end for
18: Use any of the frequency bands ticked

```

Fig. 9 Searching for the optimal frequency window

Table 3 Flattened confusion matrices resulting from the integration of several pulses, with $\sigma = 0.01$ m at $f_0 = 1$ GHz

N	Target 1			Target 2			Target 3		
	P_{cc1}	$P_{mc1,2}$	$P_{mc1,3}$	P_{cc2}	$P_{mc2,1}$	$P_{mc2,3}$	P_{cc3}	$P_{mc3,1}$	$P_{mc3,2}$
2	64.0	35.9	0.04	99.1	0.9	0	69.6	6.1	24.2
10	74.4	25.6	0	100	0	0	92.1	0.3	7.6
50	92.0	8.0	0	100	0	0	99.9	0	0.1
100	97.7	2.3	0	100	0	0	100	0	0
150	99.3	0.7	0	100	0	0	100	0	0
200	99.8	0.2	0	100	0	0	100	0	0

4.4 Integration of several pulses

4.4.1 Noiseless case: A way around the low overall probability of correct classification for the case $\sigma = 0.01$ m is to integrate over several echoes. In practice, the use of a sequential procedure to improve the performance (detection, classification) is common in radar. Using the *default* window, we make use of the results from Section 4.1.2, where we generated $M = 5000$ echoes. We assume that throughout the transmission time the targets maintain a high level of correlation. For that reason, it is important to realise that this concept requires a controlled environment. To assess the statistical properties of this approach, we repeat a Monte Carlo simulation also with 5000 runs. At every iteration, we pick N_p echoes out of the larger set and we sum their energies. We then conclude on the classification upon that compound echo. The selection of the N_p echoes is made at random so that throughout the Monte Carlo experiment different sets of echoes are used.

Finally, to smoothen the results further, for every value of N_p that we selected, we repeat our Monte Carlo experiment 50 times and average out the individual probabilities. The mean values for the probability of correct classification and misclassification ensuring this approach for different number of pulses N_p are given in Table 3 for each of the three targets.

We observe that integrating 150 pulses ensures an overall probability of correct classification $P_{cc} = 99.8\%$ in our configuration with three targets, $\sigma = 0.01$ m and $M = 5000$ realisations for each target.

4.4.2 With AWGN: When AWGN is superimposed onto the received echoes, we expect to maintain reasonable levels of performance since the noise power becomes evenly distributed on the several sub-echoes. In other words, there is no reason that it would weight more on one sub-echo than any other. We verify our

intuition with the very same Monte Carlo simulation for different values of the SNR. To keep the control of the SNR we normalise all echoes prior to adding the noise. Our findings are given in Fig. 10 for different values of N_p . We observe that reasonable values for N_p result in satisfactory values for P_{cc} under low SNR conditions, such as 0 dB or 10 dB.

5 Conclusion

We have proposed and simulated a pulse optimisation technique able to achieve high classification rates in noiseless and AWGN conditions. In our concept, the radar pulse relies on the OFDM structure and we optimise it using a genetic algorithm-based technique. Most of the classification effort is happening during the design phase of the pulse. Apart from a matched-filter which we use to localise the start of the received echo, we rely on a basic energy detector to sum the samples of the sub-echoes and decide on the target present. Remarkably, the optimisation procedure primarily concerns the magnitude of the OFDM spectrum elements (subcarriers), such that the phase can be used to reduce the PMEPR. We tested and validated our technique on a set of three synthetic targets. Each target was composed of several PEC spheres so that an analytical expression for the target reflectivity spectra could be evaluated. We discussed the bandwidth position on the frequency axis and its impact on the classification performance. We mentioned that other configurations (different target models and more targets) may infer large bandwidth and other strategies to find the optimal frequency window. We also stressed that the main challenge of this technique is to ensure a high level of correlation for the targets' RCSs throughout the bandwidth during the observation time. As a result, we recommended applying this technique under controlled environments, for example in security applications.

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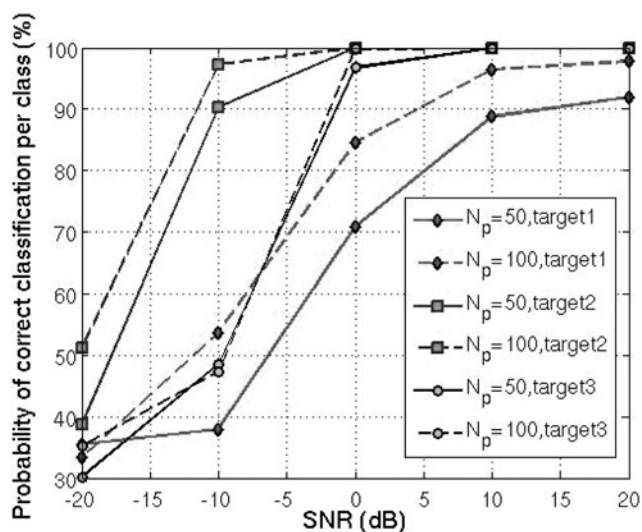


Fig. 10 Improvement resulting from the integration of several pulses on the probability of correct classification per target class for different values of SNR

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